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Amit Raj Pant

Machine Learning Engineer

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Enthusiastic Machine Learning Engineer with interest in Computer Vision, Deep Learning, Robotics, and AI for Music Understanding. Proven track record in designing end-to-end ML systems, from architecting scalable backend infrastructures using to optimizing neural networks for real-time FPGA deployment. Specialized in animal biomechanics, pose estimation, and hardware-aware neural architecture search.

Skills

Programming Languages	Python, C++, C, SQL
ML & Computer Vision	PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, OpenCV, Matplotlib, W&B
Signal Processing	Time-series analysis, Noise reduction & smoothing, FFT, Kalman Filtering, Spectral Analysis
Backend & ML Systems	Django REST Framework, FastAPI, Celery, PostgreSQL
DevOps & Tooling	Linux, Docker, Git, GitHub, Shell Scripting

Experience

Upwork, Freelancing, Deep Learning & Computer Vision Engineer — Dec 2024 – Present

Computer Vision Scientist – Equine Biomechanics PyTroch, Django REST, FastAPI, PostgreSQL, Celery — **Feb 2025 - Present**

- Owned the design and development of an end-to-end computer vision system for analyzing animal motion, from data ingestion and model training to scalable inference and deployment.
- Developed pose estimation and action recognition models, applying graph neural networks for skeletal spatial reasoning and transformer-based architectures to model long-range temporal dynamics.
- Engineered robust signal processing and preprocessing pipelines for noise mitigation, filtering, trend elimination, and normalization for downstream analysis.
- Led dataset refinement and preprocessing pipelines based on veterinary research to facilitate development of condition-specific pattern detection in animal motion.
- Developed proprietary algorithms to convert processed skeletal trajectories into intuitive statistics and insights, enabling accurate, interpretable analysis for end-users.
- Architected backend ML infrastructure using Django REST and FastAPI, managing users, projects, video data, model versions, and asynchronous inference workflows.
- Built scalable video inference pipelines with Celery and Redis, supporting background processing, progress tracking, and Dockerized production deployment.

Microscope Offset Prediction Regressor (Refactoring & Optimization) — **1 Month**

- Refactored and retrained MLP based pipeline to predict microscope offsets from image data, resolving critical data leakage issues and optimizing for TensorFlow-CUDA compatibility with multiprocessing support.
- Developed CNN architecture that improved regression performance over baseline model through systematic experimentation and enhanced codebase documentation.

LogicTronix, FPGA Design & Machine Learning Company — May 2024 – Jun 2025

Machine Learning Engineer — Part Time | PyTorch, Numpy, TensorFlow, OpenCV, C++ — **5 Months**

- Investigated and analyzed open-source pose architectures for compatibility with event-based data and DPU architecture, then developed, trained, and optimized a pose detection network capable of running in real time on FPGA hardware.
- Designed a system to automatically annotate event data via RGB conversion and fine-tune on open-source RGB pose models, increasing annotation speed by 10X.

Junior ML Engineer — Full Time | PyTorch, Numpy, TensorFlow, OpenCV, C++ — **6 Months**

- Developed a real-time FPGA-based object detection system that achieved 110 FPS on event data by designing a custom network and optimizing inference through threading, leading to 81% faster inference than conventional methods.
- Evaluated and implemented different machine learning models for identifying operational anomalies in live motor systems.
- Implemented and reviewed Spiking Neural Networks (SNNs) for event-based vision applications.

Machine Learning Acceleration — Internship | PyTorch, Numpy — **3 Months**

- Developed a 96% accurate FPGA-based real-time passenger counting system with head tracking, including dataset preparation.
- Reduced inference time of deep learning models by upto 90% through pruning, quantization, and knowledge distillation.
- Implemented and trained YOLO object detection family from scratch (YOLOv2, YOLOv3, YOLOv4, and YOLOv6) on PyTorch.

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Fusemachines — AI Fellowship **6 Months**

Capstone Project: Nepali ASR

- Developed a Nepali Automatic Speech Recognition system utilizing Whisper, capable of recognizing multiple Nepali accents.
- Collected, cleaned, and compiled a comprehensive dataset for the project. Achieved a Word Error Rate below 30.

Coursework: Data Mining, Image Processing, CNNs, RNNs, Transformers, NLP, LLMs, RL, MLops, and Deployment.

Projects

FPGA-Optimized Neural Architecture Search for CNNs to Enhance Real-time Efficiency — *GitHub* **2023 — 2024**

- Designed and innovated a new algorithm for finding efficient deep learning architectures for real-time deployment on FPGAs.
- Explored 300M architectures in MobileNetV3 search space using **FPGA-derived latency data** to guide an evolutionary algorithm.
- Demonstrated that FPGA-specific neural architecture search outperforms traditional model development methods.

LungVision: Identifying Pulmonary Disease through X-rays – *GitHub* **2023**

- Developed 87% accurate system to detect TB and Pneumonia through Lung X-rays by training ResNet and Inception networks.
- Created a dataset of lung X-rays (Normal, TB, Pneumonia) for training, and addressed class imbalance problem in the data.

EfficientNetV2 Optimization **Aug 2023**

- Pruned **EfficientNetV2's** by 88%, resulting in a model **14x smaller** and **2.5x faster**.

Data Mining Projects — *GitHub* **2023 — 2024**

- Conducted data analysis on **Go to Collage Dataset** to identify patterns affecting college enrollment and developed an **algorithm (ID3 and C4.5 from scratch)** to predict a student's probability of joining college based on these factors.
- Implemented **Principal Component Analysis (PCA)** from scratch and analyzed it on the Wheat Seed and Irish Flower dataset.
- Conducted analysis on a heart dataset to identify key risk factors, developed a 90% accurate Naïve Bayes system to identify risk.

Education

Institute of Engineering(IOE), Thapathali Campus **Sept 2019 — May 2024**

Bachelor of Computer Engineering — *79.71%* Kathmandu, Nepal

Coursework: Data Structures and Algorithms, Data Mining, Probability and Statistics, DBMS, Operating Systems, Artificial Intelligence, Computer Networks, Discrete Structure, Distributed Systems, Simulation and Modeling.

Capital College & Research Center **2017 — 2019**

High School — *3.58 CGPA* Kathmandu, Nepal

Publications

Optimizing CNN Deployment on FPGA Using Hardware-Aware Neural Architecture Search (Preprint). [PDF Link]

Honors and Achievements

Locus Winner (2024) — *Hardware Category.*
Grant Winner (2023) — *Final Year Major Project*
Docsumo Dataverse Winner (2024) — *Data Analysis*
DeerHack Hackathon Participant (2023) — *Recommendation System*

Certifications and Courses

Microdegree™ in Artificial Intelligence , Python for Everybody Specialization – *Coursera* , Machine Learning Specialization *DeepLearning.AI*, Hands-on Introduction to Linux Commands and Shell Scripting — *Coursera*, Introduction to Git and GitHub — *Google, Coursera*

Hobbies

Playing Music, Drumming, Chess, Reading Books, Offroad Motorbiking

† Link to all Certificates